



Effect of Integrated Nutrient Management on Growth, Yield and Economics of Rajma in the Subtropical Humid Zone of Assam

Babita Tamuli¹, D. Borthakur¹, S. Sarma¹, C. Thakuria¹, S. Borgohain¹, S. Paul¹

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ABSTRACT

Background: Rajma (*Phaseolus vulgaris* L.) is considered a viable crop for enhancing soil health, offering high economic returns to farmers and functioning as a short-duration catch crop. In this context, the present investigation was conducted to assess the effect of integrated nutrient management on the growth and productivity of rajma in the subtropical humid zone of Assam.

Methods: A field experiment on rajma (*Phaseolus vulgaris* L.) was conducted during two consecutive rabi seasons (2024-2025 and 2025-2026) in three villages of Dibrugarh district, Assam, involving three treatment combinations. The experiment was laid out in a randomized block design (RBD) with three replications using the variety Arun

Result: The pooled results of two years revealed that the highest growth, yield attributes and yield of rajma were recorded under treatment T₁ [NPK @ 60:45:40 kg ha⁻¹ + PSB @ 50 g kg⁻¹ seed + three foliar sprays of 2% urea at pre-flowering (45 DAS), 25% pod initiation (60 DAS) and pod development (70 DAS) stages]. Treatment T₁ recorded the maximum plant height (42.39 cm), number of pods per plant (22.26), number of seeds per pod (6.23), pod length (13.95 cm), pod width (10.64 mm), fresh pod weight (4.53 g), seed yield (17.76 q ha⁻¹) and benefit-cost ratio (2.28), which were found to be significantly superior over the other treatments.

Key words: Economics, Integrated nutrient management (INM), Rajma (*Phaseolus vulgaris* L.), Soil fertility, Yield.

INTRODUCTION

Rajma (*Phaseolus vulgaris* L.), popularly known as kidney bean, belongs to the Fabaceae family and possesses a chromosome number of 2n = 22. It is a highly self-pollinated pulse crop and is considered native to southern Mexico and Central America. In the context of rising food demand and the growing challenges posed by climate change, revitalizing the currently stagnant trend in food production has become imperative, making pulse crops increasingly important. Rajma plays a crucial role in nutritional security by serving as an affordable and valuable source of plant-derived protein, thereby contributing significantly to sustainable food systems and human nutrition.

The northeastern region of India possesses favourable agro-climatic conditions for the cultivation of rajma. Despite favourable agro-climatic conditions, the productivity of rajma remains low owing to suboptimal soil nutrient management practices. Rajma generally does not nodulate effectively with native rhizobia and therefore responds positively to higher doses of nitrogen, resulting in improved yield under balanced fertilizer and lime application (Sardana *et al.*, 2000). Long-term studies have shown that the sole application of balanced NPK fertilizers is insufficient to sustain higher productivity over time, as it may lead to deficiencies of secondary and micronutrients along with deterioration of soil physical properties (Changkiri *et al.*, 2024). Although increased fertilizer use may enhance crop production in the short term, excessive dependence on chemical fertilizers can adversely affect

¹Krishi Vigyan Kendra, Assam Agricultural University, Dibrugarh-786 010, Assam, India.

Corresponding Author: Babita Tamuli, Krishi Vigyan Kendra, Assam Agricultural University, Dibrugarh-786 010, Assam, India. Email: babita.tamuli@aau.ac.in

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soil health and sustainability in the long run. Integrated nutrient management (INM), involving the judicious combination of inorganic fertilizers, organic manures and beneficial microorganisms, is increasingly recognized as an important strategy for improving productivity, nutrient-use efficiency and soil health in pulse-based production systems. Recent studies and reviews have emphasized that the integration of biofertilizers with mineral nutrient sources can improve nutrient availability, nutrient uptake, crop productivity and economic returns while reducing the environmental risks associated with excessive or imbalanced fertilizer use (Raj *et al.*, 2023; Virk *et al.*, 2024). Biofertilizers, particularly phosphate-solubilizing microorganisms, can enhance the availability of otherwise unavailable soil phosphorus and stimulate root growth and nutrient acquisition, thereby improving the efficiency of

applied fertilizers (Virk *et al.*, 2024). Recent evidence further highlights the potential of integrated nutrient management to sustain legume productivity while improving soil fertility and reducing the environmental footprint of pulse production (Sharawat and Shilpa, 2025). Therefore, the integration of balanced mineral fertilization with microbial inoculants and supplementary nutrient application represents a promising approach for sustainable pulse production, particularly in nutrient-deficient and acidic soils.

Although the benefits of INM have been widely reported in major pulse crops, information on the combined use of balanced NPK, phosphate-solubilizing bacteria and foliar nitrogen application for rajma production under the acidic soil conditions of the subtropical humid zone of Assam remains limited. Moreover, region-specific information on the agronomic, nutrient-uptake and economic responses of rajma to such integrated nutrient management practices is scarce. Therefore, the present investigation was undertaken to evaluate the effect of integrated nutrient management on the growth, yield, nutrient uptake and economics of rajma under the subtropical humid conditions of Assam.

MATERIALS AND METHODS

The investigation was conducted in three villages of Dibrugarh district, Assam, during two consecutive *rabi* seasons, 2024-2025 and 2025-2026. The experimental sites were located at an altitude ranging from 55 to 121 m above mean sea level, between 27.33° and 27.35° N latitude and 94.96° and 95.28° E longitude. The average annual temperatures recorded at Dibrugarh during 2024-2025 and 2025-2026 were approximately 24.5°C and 23.5°C, respectively. The maximum and minimum temperatures were 34.0°C and 16.0°C, respectively, during 2024-2025 and 36.6°C and 9.5°C, respectively, during 2025-2026. The annual precipitation was 2760 mm during 2024-2025, with the highest monthly rainfall of 620.3 mm recorded in June and the average relative humidity was 80%. The annual precipitation during 2025-2026 was 2,819 mm, with the highest monthly rainfall of 478 mm recorded in July. The average relative humidity during the same period was 79%. The initial soil analysis revealed that the experimental soil was strongly acidic, with a mean pH of 5.4, organic carbon content of 1.25% and available nitrogen, phosphorus and potassium contents of 332.1, 21.81 and 219.7 kg ha⁻¹, respectively.

The experiment comprised three nutrient management treatments: T₁, recommended NPK @ 60:45:40 kg ha⁻¹ + PSB @ 50 g kg⁻¹ seed + three foliar sprays of 2% urea at the pre-flowering (45 DAS), pod initiation (60 DAS) and pod development (70 DAS) stages; T₂, recommended NPK @ 60:45:40 kg ha⁻¹; and T₃, farmers' practice involving FYM application along with an imbalanced fertilizer dose. The experiment was laid out in a Randomized Block Design (RBD) with three replications, resulting in a total of nine experimental plots. The total experimental area was

4,013.4 m², with an average gross plot area of approximately 445.93 m².

Rajma seeds were sown singly at a depth of 2-3 cm at a seed rate of 75 kg ha⁻¹, maintaining a spacing of 10 cm between plants and 30 cm between rows. Sowing was carried out between 20 and 30 November in both experimental years. The recommended dose of N, P₂O₅ and K₂O (60:45:40 kg ha⁻¹) was applied through urea, single superphosphate (SSP) and muriate of potash (MOP), respectively. The full dose of manure, phosphorus and potassium, along with half of the recommended nitrogen, was incorporated during final land preparation. The remaining half of the nitrogen was applied 30 days after sowing, as per the recommendations of Assam Agricultural University (2025).

For PSB inoculation, the seeds were treated with a PSB slurry at the rate of 50 g kg⁻¹ seed for 30 minutes and subsequently shade-dried to facilitate proper inoculation around the seed. The remaining recommended package of practices was followed for crop management. The crop was harvested when approximately 75-80% of the pods had turned brown, which occurred between 10 February and 15 March during the two experimental years. The treatments were evaluated based on growth, yield and yield-attributing parameters, soil nutrient status and economic returns.

Statistical analysis

The data recorded for the growth, yield, yield-attributing and economic parameters of rajma during the two consecutive years (2024-2025 and 2025-2026) were subjected to statistical analysis. Data from each experimental year were analyzed separately using analysis of variance (ANOVA). Prior to pooled analysis, the homogeneity of error variances between the two years was tested. Since the error variances were homogeneous and the year × treatment interaction was non-significant, the data were pooled across the two years. The pooled data from both experimental years were subjected to combined analysis of variance (ANOVA) to evaluate treatment effects. Treatment means were compared using the least significant difference (LSD) test at the 5% level of significance.

RESULTS AND DISCUSSION

Growth parameters

Application of NPK fertilizers along with PSB and three foliar sprays of 2% urea exerted a significant influence on growth and yield attributes compared to the control. Data presented in Table 1 and 2 revealed that among the three treatments, T₁ [NPK @ 60:45:40 kg ha⁻¹ + PSB @ 50 g kg⁻¹ seed + three sprays of 2% urea at pre-flowering (45 DAS), 25% pod initiation (60 DAS) and pod development (70 DAS)] recorded the highest plant height (42.39 cm), number of pods per plant (22.26), number of seeds per pod (6.23), pod length (13.95 cm) and pod width (10.64 mm).

The superior performance observed under T_1 may be attributed to the synergistic effects of balanced NPK nutrition, PSB inoculation and foliar nitrogen application. Adequate N enhanced chlorophyll formation, protein synthesis and photosynthetic activity, while P supported energy metabolism, cell division and reproductive development. Potassium improved enzyme activity, water relations and the translocation of photosynthates towards developing pods. PSB likely enhanced phosphorus availability by solubilizing otherwise unavailable soil phosphorus and may have improved root activity and nutrient acquisition. Furthermore, foliar application of urea during critical reproductive stages helped maintain nitrogen availability during pod initiation and development, thereby supporting pod growth and assimilate accumulation. The combined improvement in nutrient availability, root activity, photosynthesis and assimilate translocation consequently

resulted in greater pod length, pod weight and overall nutrient uptake.

Similar results were reported by (Sun *et al.*, 2026; Pang *et al.*, 2024; Longmatula *et al.*, 2021; Kumar *et al.*, 2009; Shwetha *et al.*, 2012; Zahida *et al.*, 2016) in French bean, as well as Verma *et al.* (2017) in soybean.

Yield and yield attributes

The integrated application of fertilizers along with PSB and three foliar sprays of 2% urea significantly enhanced yield and yield-attributing traits of Rajma compared to the control (Table 3). Among the treatments, T_1 [NPK @ 60:45:40 kg ha⁻¹ + PSB @ 50 g kg⁻¹ seed + three sprays of 2% urea at pre-flowering (45 DAS), 25% pod initiation (60 DAS) and pod development (70 DAS)] recorded the maximum fresh pod weight (4.53 g) and seed yield (17.76 q ha⁻¹). The higher yield observed under the integrated application of PSB with

Table 1: Effect of integrated nutrient management on Rajma.

Treatment	Plant height (cm)			No. of pod/plant			No. of seeds/pod		
	2024-25	2025-26	Pooled mean	2024-25	2025-26	Pooled mean	2024-25	2025-26	Pooled mean
T_1	42.99	41.78	42.39	24.87	19.64	22.26	6.71	5.75	6.23
T_2	39.20	34.24	36.72	21.95	17.92	19.94	6.03	5.70	5.87
T_3	29.28	23.19	26.24	16.15	13.53	14.84	5.71	5.03	5.37
SEm±	4.09	5.42	4.71	2.57	1.81	2.18	0.29	0.23	0.25
CD (5%)	0.85	2.37	1.51	1.70	1.55	1.45	1.49	0.72	0.62

Footnotes: T_1 : NPK@ 60:45:40 kg/ha + PSB @ 50 g/kg of seed + 3 sprays of 2 % urea at pre-flowering (45 DAS), 25% pod initiation (60 DAS) and pod development (70 DAS) stages, T_2 : N: P₂O₅: K₂O @ 60:45:40 kg/ha, T_3 : Control: Farmers' practice: FYM with imbalance dose of fertilizer.

Table 2: Effect of integrated nutrient management on Rajma.

Treatment	Length of pod (cm)			Width of pod (mm)		
	2024-25	2025-26	Pooled mean	2024-25	2025-26	Pooled mean
T_1	12.94	14.95	13.95	11.03	10.25	10.64
T_2	12.31	12.23	12.27	10.30	10.05	10.18
T_3	10.85	12.01	11.43	9.95	9.61	9.78
SEm±	0.15	0.20	0.19	0.22	0.18	0.18
CD (5%)	0.59	0.79	0.74	0.85	0.71	0.72

Footnotes: T_1 : NPK@ 60:45:40 kg/ha + PSB @ 50 g/kg of seed + 3 sprays of 2 % urea at pre-flowering (45 DAS), 25% pod initiation (60 DAS) and pod development (70 DAS) stages, T_2 : N: P₂O₅: K₂O @ 60:45:40 kg/ha, T_3 : Control: Farmers' practice: FYM with imbalance dose of fertilizer.

Table 3: Effect of integrated nutrient management on Rajma.

Treatment	Fresh weight of pod (g)			Seed yield (q ha ⁻¹)		
	2024-25	2025-26	Pooled mean	2024-25	2025-26	Pooled mean
T_1	5.58	3.47	4.53	18.01	17.50	17.76
T_2	5.57	3.30	4.44	17.05	17.25	17.15
T_3	4.33	3.10	3.72	14.95	14.50	14.73
SEm±	0.42	0.11	0.29	0.19	0.24	0.22
CD (5%)	0.77	0.71	1.0	0.67	0.82	0.76

Footnotes: T_1 : NPK@ 60:45:40 kg/ha + PSB @ 50 g/kg of seed + 3 sprays of 2 % urea at pre-flowering (45 DAS), 25% pod initiation (60 DAS) and pod development (70 DAS) stages, T_2 : N: P₂O₅: K₂O @ 60:45:40 kg/ha, T_3 : Control: Farmers' practice: FYM with imbalance dose of fertilizer.

Table 4: Effect of integrated nutrient management on nutrient status of soil after harvest of Rajma.

Treatment	Available N (kg ha ⁻¹)	Available P (kg ha ⁻¹)	Available K (kg ha ⁻¹)	Organic C (%)	Soil pH
T ₁	568.8	26.85	280.11	1.63	5.9
T ₂	478.70	26.11	277.76	1.61	5.5
T ₃	283.59	23.03	126.60	1.41	5.1
SEm±	2.09	0.67	1.2	0.08	0.17
CD (5%)	5.10	1.67	3.62	0.25	0.52

Footnotes: T₁: NPK @ 60:45:40 kg/ha + PSB @ 50 g/kg of seed + 3 sprays of 2 % urea at pre-flowering (45 DAS), 25% pod initiation (60 DAS) and pod development (70 DAS) stages, T₂: N: P₂O₅: K₂O @ 60:45:40 kg/ha, T₃: Control: Farmers' practice: FYM with imbalance dose of fertilizer.

Table 5: Effect of integrated nutrient management on economics of Rajma.

Treatment	Yield (q ha ⁻¹)	Gross income	Cost of cultivation (Rs.)	Net income (Rs.)	B:C ratio
T ₁	17.76	133200	58,500	74700	2.28
T ₂	17.15	128625	58000	70625	2.22
T ₃	14.73	110475	55900	54575	1.98

Footnotes: T₁: NPK @ 60:45:40 kg/ha + PSB @ 50 g/kg of seed + 3 sprays of 2 % urea at pre-flowering (45 DAS), 25% pod initiation (60 DAS) and pod development (70 DAS) stages, T₂: N: P₂O₅: K₂O @ 60:45:40 kg/ha, T₃: Control: Farmers' practice: FYM with imbalance dose of fertilizer.

NPK fertilizers, compared to the sole use of chemical fertilizers, may be attributed to the balanced supply of essential nutrients, improved soil health and enhanced nutrient assimilation. The yield enhancement due to PSB inoculation could be associated with increased nutrient availability in the soil and improved nodulation, which promoted better plant growth and development. This may further be linked to efficient phosphorus mobilization, greater translocation of photosynthates towards economically important plant parts and improved hormonal balance within the plant system. These findings are in agreement with the reports of Menaria and Singh (2004) in soybean, Naagar and Meena (2004) in cluster bean and Rajput and Pandey (2004) in Garden pea.

Nutrient status of soil

Treatment T₁ recorded the highest available nitrogen (568.8 kg ha⁻¹), phosphorus (26.85 kg ha⁻¹), potassium (280.11 kg ha⁻¹) and organic carbon content (1.63%) (Table 4). Soil pH across all treatments ranged from 5.1 to 5.9, with no significant differences observed among the treatments. The experimental soil was strongly acidic; a condition commonly observed in the high-rainfall regions of Assam due to the leaching of basic cations and the consequent accumulation of acidic constituents. Soil acidity can reduce phosphorus availability through fixation by iron and aluminium compounds and may adversely affect root growth and nutrient acquisition. Under such conditions, the superior performance of T₁ may be attributed to the combined effects of balanced NPK fertilization, PSB-mediated phosphorus solubilization and foliar nitrogen supplementation. The improved availability of phosphorus may have promoted root development and enhanced the acquisition and utilization of other nutrients, while foliar

urea supplied additional nitrogen during critical reproductive stages. Therefore, integrated nutrient management may be particularly effective for improving nutrient-use efficiency and productivity of rajma under the acidic soil conditions prevalent in Assam. These findings are consistent with those reported by Longkumer and Singh (2015) and Kemal *et al.* (2018).

Economics

It is evident from Table 5 that the highest cost of cultivation was recorded under treatment T₁ (₹ 58,500 ha⁻¹), followed by T₂ (₹ 58,000 ha⁻¹), whereas the lowest cost of cultivation was recorded under farmers' practice (T₃) (₹ 55,900 ha⁻¹). The highest benefit-cost ratio (B:C ratio) was observed under T₁ (2.28), followed by T₂ (2.22). The comparatively higher cost of cultivation under T₁ may be attributed to the additional expenditure incurred on PSB inoculation and foliar application of urea. Similar findings have been reported by Longmatula *et al.* (2021) and Mukherjee (2016). Moreover, the integrated nutrient management treatment (T₁) resulted in a 36.9% higher net return and a 15.15% higher B:C ratio compared with farmers' practice (T₃).

CONCLUSION

It can be concluded from the present study that Integrated Nutrient Management (INM) significantly improved the growth and yield of rajma under the subtropical humid zone of Assam, India. Among the different treatments, T₁ [NPK @ 60:45:40 kg ha⁻¹ + PSB @ 50 g kg⁻¹ seed + three foliar sprays of 2% urea at pre-flowering (45 DAS), 25% flowering/pod initiation (60 DAS) and pod development stage (70 DAS)] recorded the highest yield along with the maximum benefit-cost ratio. Therefore, T₁ proved to be the

most economically viable nutrient management practice, demonstrating the potential of this technology for enhancing productivity and supporting the socio-economic development of farmers in the region.

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Disclaimers

The views and conclusions expressed in this article are solely those of the authors and do not necessarily represent the views of their affiliated institutions. The authors are responsible for the accuracy and completeness of the information provided, but do not accept any liability for any direct or indirect losses resulting from the use of this content.

Conflict of interest

The authors declare that there are no conflicts of interest regarding the publication of this article. No funding or sponsorship influenced the design of the study, data collection, analysis, decision to publish, or preparation of the manuscript.

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